Maxwell Bay, PhD

Senior Data Scientist
Machine Learning Engineer

Contact

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Education

University of Southern California | PhD, Neuroscience
March 2022

Syracuse University | BSc, Psychology/Neuroscience
May 2011

Technical Skills

Machine Learning: Bayesian & variational inference, Gaussian processes, normalizing flows, latent-variable models & embeddings, transformers

AI Engineering: LLMs, RAG/GraphRAG, multi-agent systems & tool use, LoRA fine-tuning

Frameworks: PyTorch, JAX/Equinox, NumPyro, LangChain/LangGraph, Optuna

ML Systems: Metaflow, Airflow, Modal, Docker, CI/CD, AWS Batch, Athena, model registries

Languages: Python, R, Julia, SQL, Bash

Profile

Senior data scientist and ML engineer who leads modeling work from problem framing to nightly production. A decade building probabilistic and generative models, first in neuroscience research and now in two major league front offices. Sets technical direction across a product portfolio, develops and onboards analysts, builds reliable production systems, and translates results into decisions with executive leadership.

Experience

November 2024 – Present — Los Angeles, CA

Senior Quantitative Analyst @ Los Angeles Dodgers

- Lead analytics for the cross-departmental pitching vertical as its primary quantitative analyst, owning its interfaces with hitting, fielding and projections; partner with coaches and leadership to frame problems, set technical direction and shape key decisions
- Mentor and develop junior analysts through weekly co-development and training in baseball research; led onboarding for all new hires this year and helped recruit a top analyst; founded Research Bytes, a cross-group R&D hackathon that has produced new tools
- Built the club's batted-ball simulation engine: two chained conditional rational-quadratic spline flows (launch parameters given learned batter and pitcher latents and game context; landing point and hang time given launch and weather) with a multiclass outcome head mapped to run value (JAX/Equinox). Its Monte Carlo samples drive pitching, fielding and matchup projections and, through base-out and win-probability layers, defensive-positioning optimization, replacing direct run-value and mixed-effects models
- Designed and shipped the club's first pitcher command model, long requested but unproven, now used in coaching, projections and acquisitions: a hierarchical bivariate-normal model in which separate target and spread latents, mapped through pitch-type factor loadings, set the intended location and the log-Cholesky miss covariance (NumPyro/JAX). Globals are fit annually by amortized SVI and per-pitcher latents re-solved nightly by deterministic MAP under an AR(1) season prior, giving reproducible 20–80 grades across MLB and MiLB
- Own the quantitative analysis team's pitching products and core hitting models end to end, including retraining across dependent models (Metaflow, AWS Batch, Modal, Airflow, point-in-time model registry): the pitch grader models, a binary tree of Bayesian Gaussian-process classifiers and a GP regressor on batted-ball value with factored pitcher and batter effects, rolled up to run-expectancy deltas; and the contact-quality models, including a forward-looking variant pretrained on run value and fine-tuned on split-half, out-of-sample targets
- Developing the club's next-generation pitch grader and pitch-level game simulator: a generative model of the plate appearance in which a causal transformer over the pitch sequence conditions a pitch law (type, kinematic and location spline flows) and a matchup outcome tree ending in a launch-spec flow, with pitcher, batter, catcher and umpire latents; pitch value, actor-neutral quality and Stuff are integrals of the model (JAX/Equinox)
- Built a model that predicts which new pitches a pitcher can develop, from amateur delivery to pro arsenal up to four seasons out: a latent neural point process with a set-transformer encoder over pitch, delivery and kinematic tokens and a decoder pairing a Poisson count head with a conditional rational-quadratic spline flow and Gibbs repulsion; beats linear-Gaussian and kNN baselines by about 0.6 nats per point in held-out log score (JAX, FlowJAX, Modal)

April 2024 – November 2024 — Los Angeles, CA

Senior Data Scientist/AI Engineer @ PortfolioT

- Led data science and AI strategy for incubator and venture-studio startups
- Built GraphRAG chat systems that ground LLM responses in current information: knowledge graphs drafted by an LLM agent swarm that cross-checked its extractions in a LangGraph review cycle before human validation, served by hybrid vector retrieval and graph traversal in Neo4j behind a model-agnostic LLM layer
- Engineered autonomous multi-agent research systems in LangGraph that synthesize internal documents, social media and multimodal content: crawler agents read and filter web sources for researcher agents, while a liaison agent directs the next round of search from the researchers' open questions
- Built conversational digital twins by fine-tuning per-person LoRA adapters on open-weight Llama with volunteers' transcribed speech, with an automated evaluation framework scoring each twin's fidelity to its source

Experience continues on page 2

Experience (continued)

November 2021 – April 2024 — Houston, TX

R&D Analyst @ Houston Astros

- Delivered models, tooling and infrastructure behind decisions across baseball operations, partnering with major league coaches on metric design and engineers on production
- Built the club's pitch grader, used from coaching through player evaluation: a multiclass neural classifier over the terminal nodes of the pitch-outcome tree and a mixture density network on batted-ball run value, conditioned on trajectory physics and context (PyTorch)
- Designed the organization's first batter pitch-recognition model, fully analytic Bayesian, which flagged pitchers whose pitch types were too easily recognized: 9-parameter fits rotated into the batter's gaze frame, per-pitcher multivariate-normal priors over the first 180 ms of flight, normal-normal conjugate updates under observer noise for pitch-type posteriors and zone probability, and a learned early, on-time or late timing emission
- Fit partially pooled player effects on swing-start and commit times as functions of pitch velocity, break and plate location (hierarchical Bayesian regression, SVI in NumPyro), then mapped start-time deviations to on-field outcomes

August 2015 – March 2022 — Los Angeles, CA

PhD Candidate/Researcher @ Broad Center for Regenerative Medicine, University of Southern California (Mentor: Dr. Michael A. Bonaguidi)

- Oversaw the lab's computational wing: set bioinformatics and data science direction, mentored undergraduate and MS students, and advised other labs on statistics
- Designed a pipeline that nominates pharmaceutical and genetic interventions for diseased tissue from single-cell RNA-seq: autoencoder dimensionality reduction, healthy-versus-diseased expression trajectories, and signature reversal against Connectivity Map perturbation profiles to rank perturbagens that move diseased expression toward the healthy state (R)

August 2011 – August 2015 — Los Angeles, CA

Programmer Analyst @ Laboratory of NeuroImaging, USC

- Built image segmentation and visualization tools for mouse-brain connectomics in Alzheimer's, Huntington's and autism disease models